

On the use of instrumental data for infragravity wave simulations

Joao Albuquerque¹, Simon Weppe¹ and Alexis Berthot¹

¹ MetOcean Solutions, Raglan, New Zealand; j.dealbuquerque@metocean.co.nz

Abstract

Infragravity (IG) waves are surface ocean waves with periods ranging from 25 s to 300 s. They typically have low amplitudes and wavelengths an order of magnitude longer than wind-generated waves. IG waves can penetrate ports and harbours, and their reflection off port's internal walls and structures can generate resonance when the port's basin and IG wave lengths are similar. The amplification of the IG wave energy by resonance can cause surge motion on moored vessels, which can break mooring lines and threaten personnel safety.

Most studies considering IG wave effects in harbours use white noise (synthetic) boundary forcings in simulations to identify the wave periods that can cause resonance within a port layout, or in run spectral simulations based on the statistics of the local wave climate. Although synthetic boundaries can effectively provide such resonance periods for ports, it is not possible to predict whether these long-period oscillations or resonance will effectively occur under the natural wave climate. On the other hand, simulations based on the local wave climate use bulk wave parameters, which are not capable to represent the multiple wave systems (i.e., seas and swells) that can be present on the wave spectrum.

In this paper, we use concurrent offshore and in-basin wave measurements to study IG waves processes in Eastland Port, Gisborne (NZ). We selected a range of IG wave events recorded by an IG wave radar within the port basin and used the simultaneous spectral offshore wave condition as boundary forcings for SWASH simulations. Results show the different spatial distributions of IG wave heights that can occur within the port basin, and identifies the differences in wave spectra for the cases modelled. The study highlights the importance of using realistic wave conditions for IG wave investigations.

Keywords: wave buoy, infragravity waves, SWASH, ports, harbours.

1. Introduction

Wave energy present in groups of multiple waves can be released as an infragravity (IG) wave as they propagate shoreward and interact with the seabed. IG waves are typically lower than 0.01 m in deep water (Webb et al., 1991) and have very long wavelengths, with periods ranging from 25 s to 300 s. Like ordinary waves, IG (or long) waves can penetrate ports and harbours, however as IG waves reflect off the port's internal walls and structures, they can generate resonance when the port's basin and IG wave lengths are similar. The amplification of IG wave energy by resonance can cause long-wave induced surge motion on moored vessels, which can break mooring lines and threaten personnel safety. Understanding the long waves behaviour as they penetrate harbours is therefore fundamental for the design of ports and other nearshore structures, as well as for the development of IG wave forecasts that minimise downtime of port operations without compromising the port's crew safety.

A number of authors have studied the mechanisms that generate long waves, both on the offshore (Biesel, 1952; Longuet-Higgins & Stewart, 1962) and the nearshore (Battjes et al., 2004; Janssen et al., 2003; Masselink, 1995) areas. On the offshore areas, long waves are formed through nonlinear interactions between different short wave systems at discrete frequencies as they propagate shoreward. This long wave propagates in phase

with the wave group and therefore is called a bound wave (Longuet-Higgins & Stewart, 1962). Along the nearshore areas, mechanisms such as moving breakpoint (List, 1991; Symonds et al., 1982) and bore merging (Bradshaw, 1980; Huntley & Bowen, 1975; Mase & Iwagaki, 1984) can also generate IG waves within the surf zone.

A wealth of research using phase resolving models to address the IG wave effects in harbours is available on the literature with different types of boundary forcings. Examples of boundary forcings used include monochromatic waves (Costas et al., 2022; Gao et al., 2016; Kumar & Gulshan, 2017), hindcast data (Guo et al., 2021; Panigrahi et al., 2015), and instrumental data such as wave buoys (Bellafont et al., 2018) and pressure sensors (Kwak et al., 2020). Despite the contributions of these studies to the understanding of infragravity waves, each type of boundary forcings used have inherent limitations. Monochromatic waves can be very effective at providing resonance periods in ports, but it is not possible to predict whether these long-period oscillations will occur under the natural wave climate. Hindcasts can provide long term statistics of wave climate for a given area, from which statistically sound wave conditions can be modelled, however either bulk or spectral wave parameters estimated by hindcasts are not as accurate as in situ measurements. Finally, model simulations forced by bulk wave parameters from instruments also lack the information of the multiple

wave systems that can be present on a sea state, which is critical for the proper generation of IG waves since they depend on nonlinear interactions between wave systems with different frequencies.

Hence, studies outlining the different results from IG wave simulations from these different boundary forcings (i.e., monochromatic, hindcast bulk and spectral parameters, as well as buoy bulk and spectral parameters) can add to our understanding on how the interaction of the different wave systems present on a sea state can change the resulting IG wave inside a port basin. Such understanding has the potential to improve the accuracy of IG wave forecasts, allowing for reduced port downtime and increased safety.

In the present paper we used an IG wave radar to select specific long wave events in Eastland Port, Gisborne. For each event, we compared the results of a SWASH¹ simulation forced by the full wave spectrum to the results of a simulation forced by bulk wave parameters (both forcings derived from a wave buoy near the offshore boundary). Results show the different spectral shapes and spatial distributions of IG waves that can occur at the boundary and within the port basin for the different cases simulated and reinforces the importance of the use of realistic wave conditions for such studies.

2. Data and model

2.1 Bathymetry

The bathymetry used for the simulation domain and port layout (Figure 1) was interpolated from a merged bathymetric dataset provided by Eastland Port. All datasets were converted to Mean Sea Level (MSL) considering $MSL = +1.056$ m LAT (Lowest Astronomical Tide or Chart Datum).

2.2 Wave data

Two wave measuring instruments are available at Eastland Port, the first is a Triaxys offshore wave buoy (WB) and the second is a wave radar (WR), installed inside the port. The geographic location of both devices is presented in Table 1.

Table 1 Depth and geographic location of the instruments used in this study. WB and WR stand for wave buoy and wave radar respectively.

	WB	WR
Depth	19	n/a
Easting (m)	2035814.0	2037343.7
Northing (m)	5704860.9	5707439.2

The WB is moored about 2.5 Km offshore of the harbour entrance, at approximately 19 m water depth as per Figure 1 (top). The 2D wave spectrum

provided by the buoy has frequency and directional domains ranging from 0.03 to 0.635 Hz (frequency spacing of 0.005 Hz), and 0 to 360° (directional spacing of 3°) respectively. Additional (bulk) wave parameters provided by the buoy include significant wave height (H_s), peak wave period (T_p), mean wave direction (θ_m) and directional spread (σ_θ). Both spectral and bulk buoy measurements are available at a 20-minute temporal resolution.

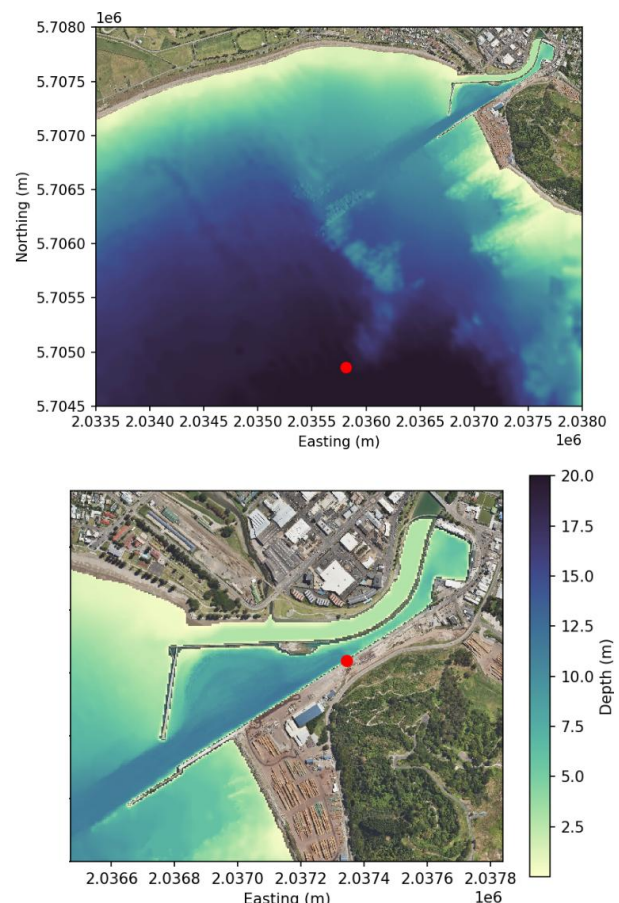


Figure 1 Bathymetry of our SWASH model domain and wave buoy (WB) location (top), Eastland port basin bathymetry and wave radar (WR) location (bottom). Red dots represent instrument location, colours indicate the water depth as per colour scale.

The WR, located at the port's constriction (Figure 1, bottom), provides the water level and a range of non-directional wave parameters for both infragravity and short waves in a 1-minute temporal resolution. Parameters of waves are measured for wind sea waves, i.e., waves with periods of up to 7 s, swell waves, or waves with periods within 7 and 25 s, and the total (wind sea + swells) waves within this period range. For infragravity waves, parameters are calculated over waves with periods larger than 25 s and waves whose periods are between 25 and 120 s.

¹ swash.sourceforge.net/

A time-series plot with both offshore and in-basin wave measurements, as well as water level, illustrates the dynamics of such parameters in our domain (Figure 2).

2.3 Model domain

We set up our SWASH model with a domain of 900x700 grid cells with a spatial resolution of 5.0 m, which extends from (NZTM) 2033500 to 2038000 m East and from 5704500 to 5708000 m North (see Figure 1, top).

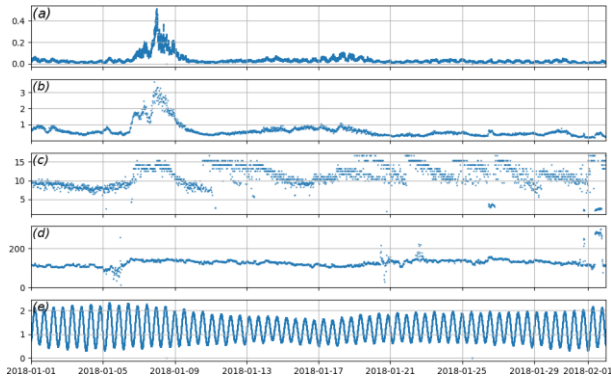


Figure 2 Time series of recorded wave parameters at Eastland Port, infragravity wave height (a) and offshore significant wave height (b), peak wave period (c), mean magnetic wave direction (d) and tide level (e).

2.4 Boundary forcings

Spectral and bulk wave data from the WB were applied as boundary forcings in our simulations. Despite the different geographic location of the wave buoy and the boundaries, we assume that the changes in the wave characteristics are negligible over this short distance.

Two IG wave height events were selected to our SWASH simulations. The first event's IG wave height (H_{SIG}) lies within the limits of safe port operations, i.e., $H_{SIG} = 0.2$ m, whereas the second event's IG wave height is of 0.4 m (lg20 and lg40 events hereinafter). In order to define the forcings of our simulations, first we selected WR H_{SIG} measurements that: (i) are within ± 0.005 m of each targeted IG wave height value and (ii) occurred during a tide height of 2.09 ± 0.02 m CD. Next, we found the closest WB reading (in time) of each selected radar measurement in the previous step, corrected the magnetic declination of the instrument, and used its data as boundary forcing for our SWASH simulations. We show the details of the offshore wave conditions used in Table 2.

Table 2 Details of the boundary forcings used in our SWASH simulations.

	lg20	lg40
Time (NZST)	20150719 08:20	20150615 04:00
H_s (m)	2.18	3.12
T_p (s)	16.7	15.4
θ_m (°)	164.5	161.0
σ_θ (°)	26.4	14.3

Each SWASH IG simulation was forced in two different ways: the first used the full 2d spectral buoy readings and the second used a generic JONSWAP spectrum reconstructed from the buoy bulk parameters of H_s , T_p , θ_m and σ_θ (measured and reconstructed spectrum simulations hereinafter).

2.5 Model setup

The model was run in depth-averaged mode, using the hydrostatic front approximation (Smit, P. et al., 2013) to account for wave breaking. A 500 m wide (~4-5 wave lengths) sponge layer was added to the western end of the model domain, allowing waves to propagate through the model's western boundary without reflection. The bottom friction values applied along the domain were of 0.025 over areas of steep reefs and 0.019 on the remaining areas (Figure 3).

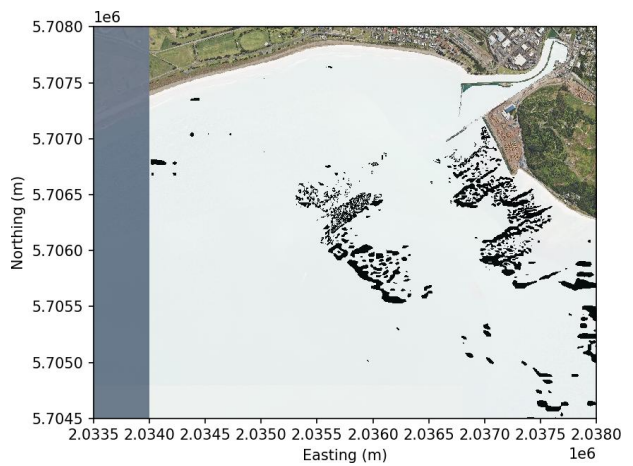


Figure 3 SWASH domain overlaid with (i) a sponge layer (grey), (ii) steep (reef) areas with friction of 0.025 (black) and areas with friction of 0.019 (white).

The different structures and revetments present in the port interact (i.e., reflect or absorb) the wave energy in different ways. This behaviour was represented in SWASH through “porosity layers” (SWASH team, 2018). A porosity layer uses a porous flow model with Van Gent (1995) equations to obtain the amount of wave/flow transmitted past an obstacle (SWASH team, 2018). In this context, the porosity values range from 0.0 to 1.0, where a porosity value of 0.0 represents an impermeable wall (i.e., no wave transmission) and a porosity of 1.0 allows for full wave transmission (i.e., “wet” point).

The location and type of revetment of each port's structure is shown in Figure 4. The rock/concrete unit revetments were assumed to have a porosity of 0.45 as recommended in SWASH team (2018). The porosity of the eastern end of the southern wall was set to a similar value to account for the presence of large blocks that separated from the main structure and were visible in the multi-beam surveys. A value of 0.0 was used for walls (i.e. fully reflective), and a

generic value of 0.65 was set for the piled decks inside the port.

2.6 Output and postprocessing

We ran each simulation for 3-hours, generating two types of outputs of sea surface elevation. The first were point outputs at the WB and WR locations, and the second was a gridded output over the port's basin area. The sea surface elevation values at the WB and WR locations were low-pass filtered using the Fourier transform to select the frequency bands within [25 – 250] seconds. Next, we calculated the infragravity wave height by integrating the wave spectra over each correspondent (filtered) frequency band. For the gridded outputs, the wave height parameters were calculated as $4\sigma(se)$, where se is the sea surface elevation and σ is the standard deviation.



Figure 4 Location of the porosity layers employed to reproduce the different structure types and revetments inside Eastland Port.

3. Results and discussion

3.1 Point output comparison

The values of H_{SIG} calculated at the WR output show better agreement between the radar readings and measured spectrum simulations compared to the reconstructed spectrum simulations (Table 3). In particular, the largest H_{SIG} difference, 0.1 m (45%) occurs for the smallest IG event (Ig20). For the same event and the spectral simulation, the difference is of 0.02 m (9%).

Table 3 H_{SIG} values of our SWASH measured and reconstructed spectral simulations at WR.

	Ig20	Ig40
Measured (m)	0.22	0.43
Reconstructed (m)	0.32	0.48
Diff measured (m)	0.02	0.03
Diff reconstructed (m)	0.10	0.08

The differences between the H_{SIG} values of each simulation (i.e., measured or reconstructed spectra) are explained by the energy peaks seen between the 0.04 and the 0.004 frequency bands (25 to 250 s) in Figure 5. For the measured spectra simulation (Figure 5, top), the peaks within the IG

frequency band are much lower than the peaks of the reconstructed spectra simulation (Figure 5, bottom). Another noticeable feature is the presence of energy in frequency bands smaller than 0.004 Hz in the measured simulation and nearly no energy in these bands for the reconstructed simulation. The same pattern of energy distribution within the IG band occurs between the measured and reconstructed runs of the Ig40 event (not shown).

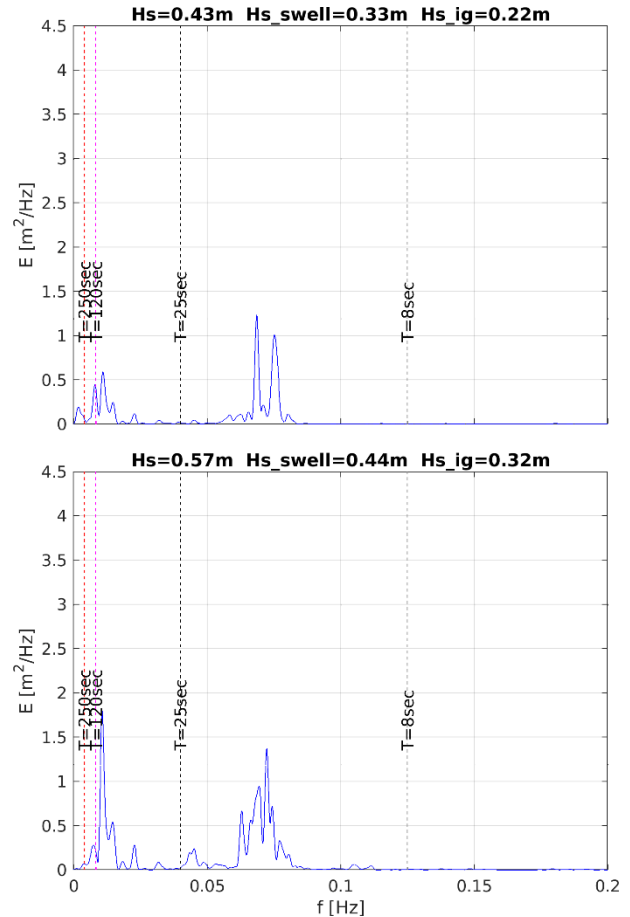


Figure 5 One-dimensional wave spectrum of the Ig20 event with measured (top) and reconstructed (bottom) forcings at WR.

Further investigation on the difference between the two Ig20 events highlights that the generic two-dimensional wave spectrum reconstructed from the bulk wave parameters may not always be a proper representation of the original wave spectrum. The one-dimensional wave spectrum calculated at the WB location shows different energy peaks within the swell and wind sea frequencies for the measured and reconstructed spectral simulations.

While the measured spectrum simulation plot (Figure 6, top) shows three energy peaks within the swell frequency range (0.04 to 0.125) and one energy peak in the wind sea frequency range (> 0.125), the reconstructed spectrum simulation plot (Figure 6, bottom) expectedly shows a typical unimodal wave spectrum with a much larger energy

peak and most of its energy in the swell frequency band.

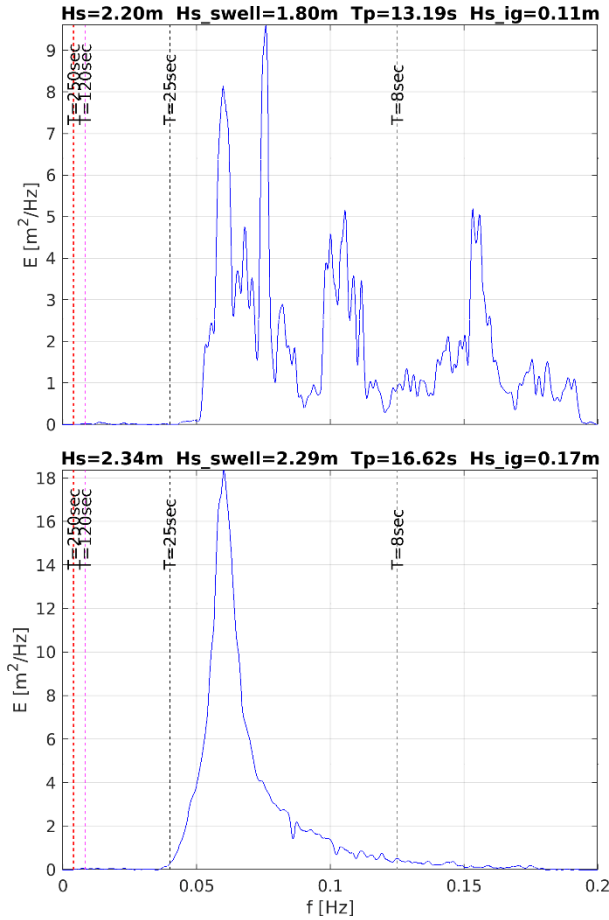


Figure 6 One-dimensional wave spectrum of the Ig20 event with measured (top) and reconstructed (bottom) spectral forcings at WB.

Such variation of the spectral shapes reflects directly on the wave parameters calculated for each spectrum. For instance, the differences between values of H_s , swell wave height (H_{sSWL}), T_p and H_{sIG} of the two Ig20 simulations are 0.14 m, 0.49 m, 3.42 s and 0.16 m respectively. The values of these wave parameters for the all the simulations at WR are shown in Table 4.

Table 4 Bulk wave parameters calculated from the wave spectrum at WB for the measured (m) and reconstructed (r) IG events simulation.

Event	H_s (m)	H_{sSWL} (m)	T_p (s)	H_{sIG} (m)
Ig20 (m)	2.20	1.80	13.20	0.11
Ig20 (r)	2.34	2.29	16.62	0.17
Ig40 (m)	3.03	2.98	15.32	0.24
Ig40 (r)	3.03	2.95	15.24	0.22

Different spectral shapes also occur in the Ig40 spectral plots at WB (not shown), however, as the two spectra are more similar in the Ig40 case, the differences between the wave parameters are not as large as for Ig20 (see Table 4).

3.2 Gridded outputs

Maps of IG wave heights of the Ig20 event show lower values of H_{sIG} for the measured spectrum simulation (Figure 7 top) compared to the reconstructed spectrum simulation (Figure 7 bottom). In general, the spatial distribution of IG wave heights is similar between the two events, with the peak of H_{sIG} located slightly northeast of the basin's restriction, and the troughs in the wider parts of the port (i.e., turning basin and the marinas at the port's north-eastern end).

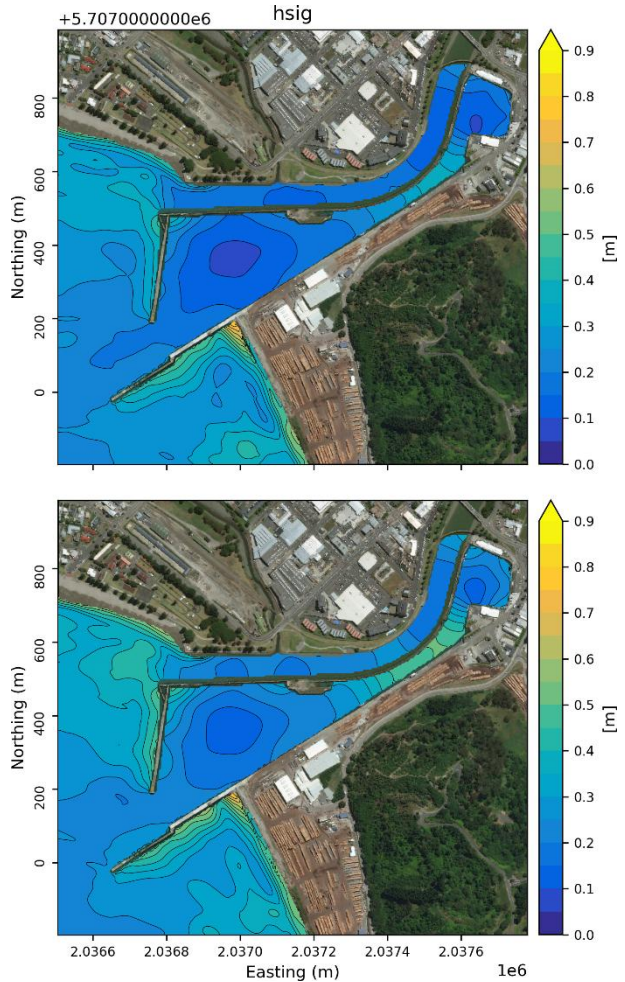


Figure 7 Spatial H_{sIG} inside Eastland Port's basin for the Ig20 event with measured (top) and reconstructed (bottom) spectral forcings.

The spatial differences in H_{sIG} between the two Ig20 simulations are shown in Figure 8. Inside the port's basin, we see an area of slightly smaller H_{sIG} for the reconstructed spectrum simulation, as well as two areas of larger H_{sIG} , adjacent to the port's basin constriction. These areas of smaller and larger H_{sIG} show the higher amplitude of H_{sIG} of the Ig20 reconstructed simulation.

The two Ig40 simulations show more similar one-dimensional spectrum at WB (not shown), which also reflects on their bulk wave parameters, as per Table 4. However, the H_{sIG} distribution inside the

port's basin is remarkably different between the measured and reconstructed spectra simulations.

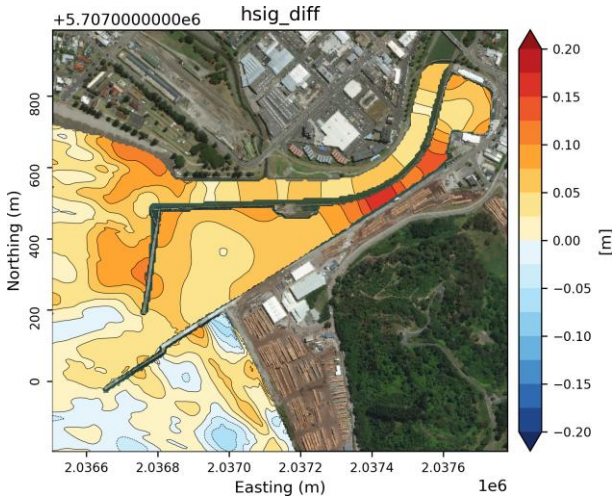


Figure 8 H_{sig} difference between measured and reconstructed forced runs of the Ig20 event. Red colours show reconstructed forced H_{sig} larger than measured forced H_{sig} , blue colours show reconstructed forced H_{sig} smaller than measured forced H_{sig} .

The measured spectrum simulation of the Ig40 event (Figure 9, top) show a large peak of H_{sig} northeast of the port's constriction and a long area with a flatter gradient southeast of the constriction. For the reconstructed simulation (Figure 9, bottom), the peak of H_{sig} is smaller, however it is located at the narrowest point of the constriction, which increases the IG gradient near the WR. These differences are more evident in Figure 10, where difference gradients suggest IG waves of different lengths.

These IG waves are generated as incoming wave systems propagate over the relatively shallow bathymetry and reefs, transferring their energy to lower frequencies through nonlinear interactions. The different wave energy distributions across the wind sea and swell frequencies for each simulation eventually cause different patterns of energy transfer to the lower frequencies, resulting in different IG wave heights within the port basin.

As these IG waves penetrate the port basin, they interact with its internal structures, which can attenuate or amplify specific infragravity frequencies depending on the wave and port's characteristics. Therefore, a broader analysis of the wave spectrum as it propagates through the domain and inside the port is suggested as future study.

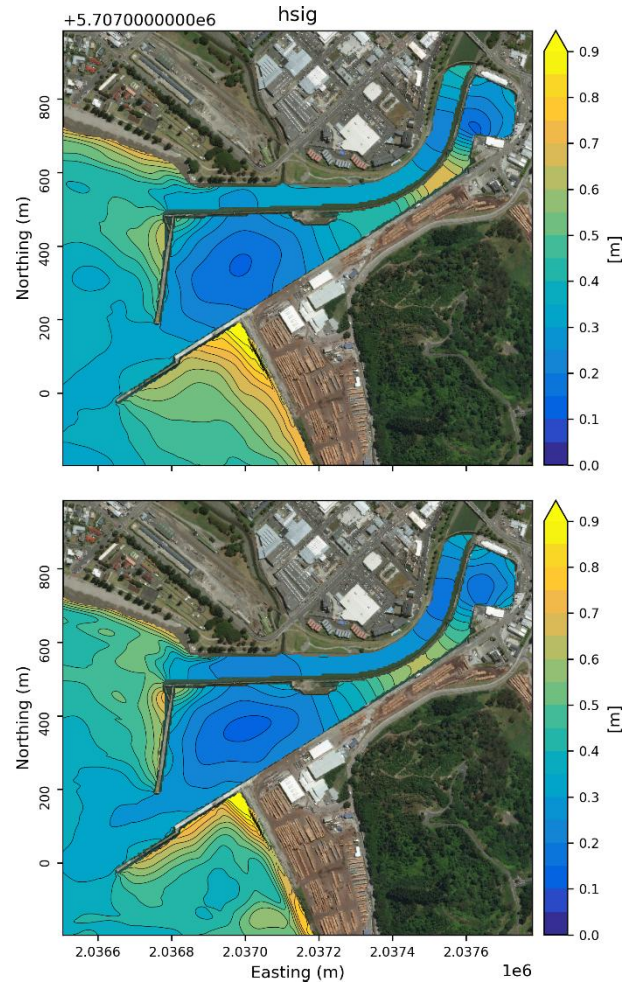


Figure 9 Spatial H_{sig} inside Eastland Port's basin for the Ig40 event with measured (top) and reconstructed (bottom) spectral forcings.

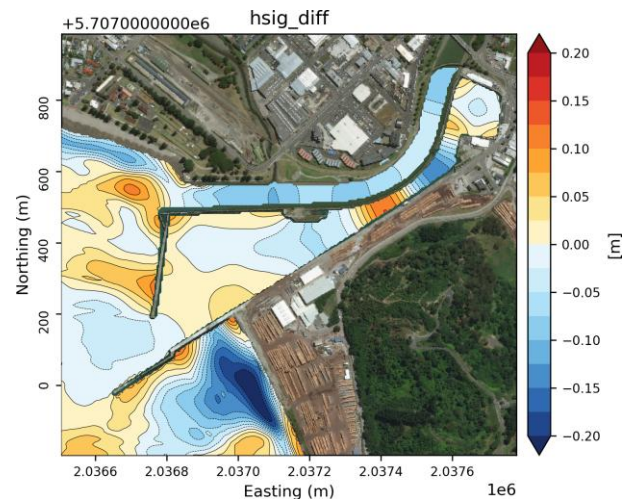


Figure 10 H_{sig} difference between measured and reconstructed forced runs of the Ig20 event. Red colours show reconstructed forced H_{sig} larger than measured forced H_{sig} , blue colours show reconstructed forced H_{sig} smaller than measured forced H_{sig} .

4. Conclusion

In the present work, we studied how the full 2d spectrum measured by an offshore wave buoy compares with the reconstructed spectrum from the buoy's bulk wave parameters in a simulation of IG waves at Eastland Port, Gisborne. Different IG wave events were simulated with the SWASH wave model and their results were compared to measurements at a wave radar inside the port.

For the cases studied, measured spectrum forced simulations show better agreement with the wave radar measured H_{SIG} compared to the reconstructed spectrum forced simulations.

An analysis of the one-dimensional wave spectrum extracted at the wave radar location showed that measured spectrum simulations have lower energy peaks in the IG wave frequency bands and slightly more energy in frequencies lower than IG wave band. Further analysis of one-dimensional wave spectra at the buoy location identified the different shapes of the spectrum reconstructed from bulk wave parameters and the spectrum measured by the wave buoy.

Comparison of maps of IG wave heights for each IG event simulation showed the different gradients of H_{SIG} present inside the port's basin. Despite the difference in the spectral shape of the Ig20 simulations at the wave buoy, the IG gradients observed inside the basin were relatively similar. In contrast, the Ig40 event has a relatively similar wave spectrum between the measured and the reconstructed spectrum at the WB location, and its simulation resulted in remarkably different spatial patterns of H_{SIG} inside the port's basin.

The differences in spatial patterns of H_{SIG} found in the port's basin are likely to be caused by the nonlinear interactions of the different wind sea and swell wave systems present at each simulated event.

5. References

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